

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
CO2_AT1 — CHART CONSTRUCTION TEST

Course Code:	DSA06	Course Title:	Data Handling and Visualization
CO Assessed:	CO2 – Construction (BL3)	Assessment:	Assessment Tool 1 – Chart Construction
Weightage:	20% of CIA	Total Marks:	25 (5 Questions × 5 Marks)
CO2 Statement:	<i>Apply appropriate chart construction techniques — correct chart type, axes, scales, and calculations — to accurately represent given data.(BL3)</i>		
Student Name:	JOTHIKA M	Reg. No.:	192324235
Section / Batch:	SLOT C	Date:	30-07-2026

Code of Conduct

I JOTHIKA M(192324235) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: *Jothika M*

Instructions

- This test contains 5 questions, each carrying 5 marks. Total: 25 Marks.
- No negative marking. Unanswered questions carry zero marks.
- No calculators or electronic devices permitted.
- Duration: 75 minutes.
- Graph paper or a ruler is recommended for accurate, to-scale drawing.
- Show all calculations (e.g., percentages, angles, bin ranges, five-number summaries) as part of your answer.
- Every chart must include a title, correctly labeled axes, and a legend where applicable.

Annexure A – CHART CONSTRUCTION

Problem 7 — Pie Chart — Angle Calculation (5 Marks)

Data: Market share: Company A = 40%, Company B = 25%, Company C = 20%, Company D = 15%.

Task: Construct a pie chart representing this market share. Show your calculation of the angle (in degrees) for each company's slice before drawing it.

Problem 8 — Pie Chart — From Raw Counts (5 Marks)

Data: Survey responses (200 total): Yes = 110, No = 60, Undecided = 30.

Task: Convert each count to a percentage, then construct a pie chart with correctly calculated slice angles for each response category.

Problem 9 — Histogram — Binning (5 Marks)

Data: Exam scores for 20 students: 45, 52, 58, 61, 63, 65, 67, 68, 70, 72, 73, 75, 76, 78, 80, 82, 85, 88, 90, 95.

Task: Construct a histogram using 5 equal-width bins spanning the full data range. Show your bin boundaries and the frequency count in each bin.

Problem 10 — Scatter Plot — Basic Construction (5 Marks)

Data: Study hours vs. exam score for 8 students: (2,50), (3,55), (4,62), (5,68), (6,74), (7,80), (8,85), (9,90).

Task: Construct a scatter plot plotting each (x, y) pair as an individual point. Label both axes with appropriate scales.

Problem 11 — Scatter Plot — With Trend (5 Marks)

Data: Advertising spend (\$'000) vs. sales (\$'000) for 10 companies: (5,20), (8,28), (10,35), (12,40), (15,50), (18,55), (20,62), (22,68), (25,75), (28,82).

Task: Construct a scatter plot of this data and sketch an approximate trend line through the points, describing the relationship you observe.

Rubrics for Calculation **ASSESSMENT RUBRIC** AT1 — Chart Construction Test

CO2 · BT3: Apply ·

This rubric is used to assess student responses to the CO2-AT1 Chart Construction Test, in which students are given a small dataset and must construct an accurate, correctly labeled chart of a specified type.

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25%	Demonstrates thorough, accurate understanding of the correct chart type, and the axes, scale, and elements required to construct it for the given data.	Good understanding of the required chart type and construction elements, with only minor gaps.	Basic understanding of chart construction, but with some misconceptions about the correct chart type, axes, or scale.	Poor understanding of core construction concepts; selects a chart type or elements fundamentally unsuited to the data.
Explanation	25%	Shows detailed, clearly worked calculations (e.g., angles, bin ranges, five-number summary) and explains each construction step.	Shows clear calculations and construction steps, with adequate explanation for most parts.	Calculations or construction steps are shown but incomplete, or explanations are superficial.	Little or no working shown; calculations and construction steps are missing or unexplained.
Application	20%	Effectively applies chart construction techniques to produce a complete, accurate chart correctly matched to the data.	Applies construction techniques to produce a correct chart, with only minor errors.	Limited application of construction techniques; the chart partially represents the data but has notable errors.	Fails to apply appropriate construction techniques; the chart does not correctly represent the data.
Organization	15%	The chart is neatly and logically constructed, with a title, correctly labeled axes, a legend where applicable, and elements in a clear order.	The chart is well organized, with only minor issues in labeling, legend, or layout.	Basic construction; required elements (title, axes, legend) are present but poorly arranged or partly missing.	Poorly constructed; the chart is missing key elements such as a title, axis labels, or a legend.
Accuracy	10%	The chart is	Mostly	Partially correct;	Largely

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
		completely accurate — all values, proportions, scales, and calculations correctly match the given data.	accurate, with only minor omissions or a small error in one value or calculation.	some plotted values, proportions, or calculations are missing or incorrect.	inaccurate; major errors in plotted values, proportions, or calculations relative to the given data.

Scoring Guide

For each criterion, award a performance level (1–4) based on the descriptors above. Multiply the level achieved (as a percentage of 4, i.e., Level 4 = 100%, Level 3 = 75%, Level 2 = 50%, Level 1 = 25%) by the criterion's weightage, then sum across all five criteria to obtain the total score out of 100%.

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

CO 2 AT 1

CHART CONSTRUCTION TEST

Problem 7 – Pie Chart – Angle Calculation (5 Marks)

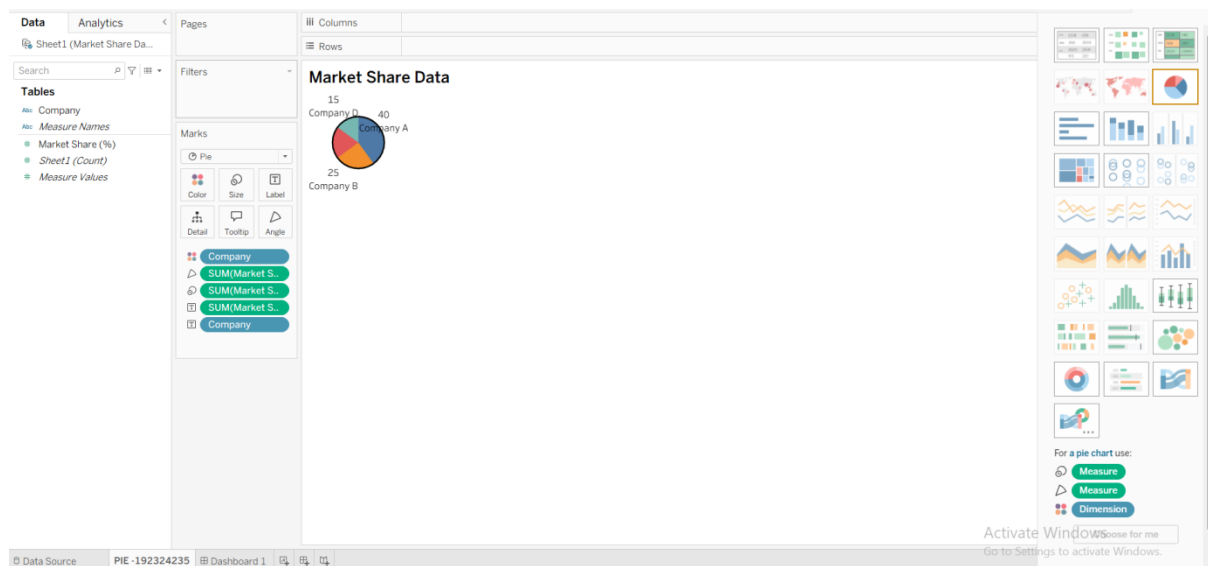
Introduction

A pie chart is a circular statistical graph used to represent parts of a whole. Each category is shown as a slice of the circle, where the size of each slice is proportional to its percentage of the total. Since a circle contains 360°, the angle of each slice is calculated using the formula:

$$\text{Angle} = (\text{Percentage} \div 100) \times 360^\circ$$

Market Share Data

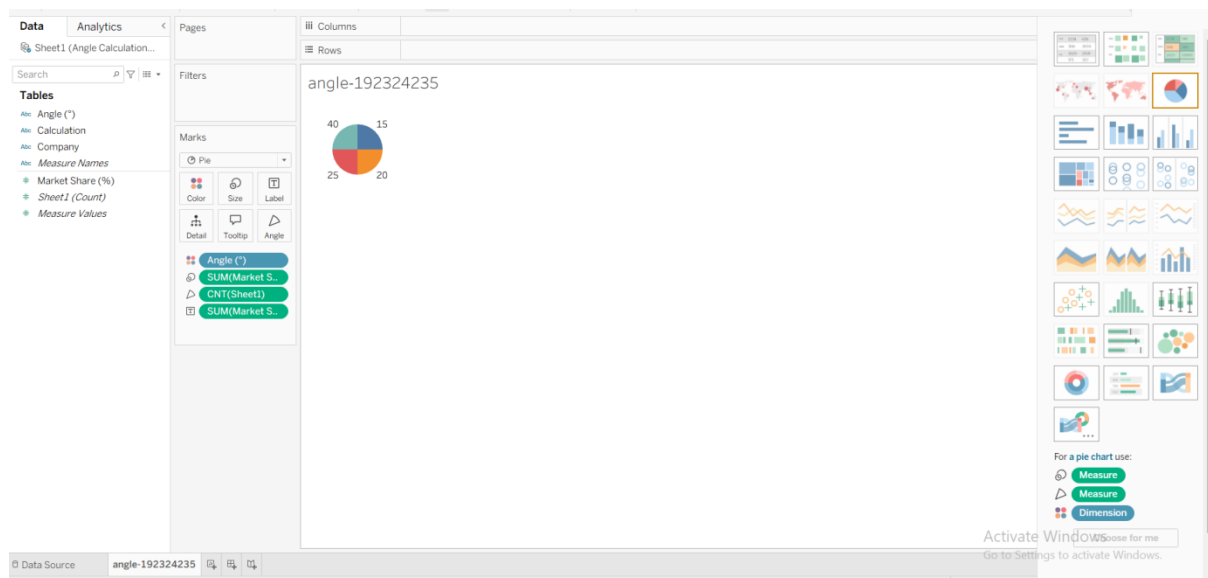
Company	Market Share (%)
Company A	40
Company B	25
Company C	20
Company D	15



Angle Calculations

Company	Market Share (%)	Calculation	Angle (°)
Company A	40	$(40/100) \times 360$	144°
Company B	25	$(25/100) \times 360$	90°
Company C	20	$(20/100) \times 360$	72°
Company D	15	$(15/100) \times 360$	54°

$$\text{Total Angle} = 144^\circ + 90^\circ + 72^\circ + 54^\circ = 360^\circ$$



Evidence

The pie chart shows that Company A has the largest market share (40%) and therefore occupies the largest slice (144°). Company D has the smallest market share (15%) with the smallest slice (54°). Together, the four slices represent the entire market (100% or 360°).

Interpretation

The pie chart clearly compares the relative market share of each company. Company A dominates the market with nearly half of the total share, while Companies B and C maintain moderate market positions. Company D contributes the smallest portion, indicating comparatively lower market presence.

Conclusion

A pie chart effectively displays proportional data by representing each company's market share as a portion of a circle. Calculating slice angles ensures accurate visualization and helps compare the contribution of each company to the overall market.

Problem 8 – Pie Chart – From Raw Counts (5 Marks)

Introduction

A pie chart can also be created from raw frequency data. The raw counts are first converted into percentages, and then each percentage is converted into an angle out of 360° . This allows the survey results to be represented visually as portions of a complete circle.

Survey Data

Response	Count
Yes	110
No	60
Undecided	30
Total	200

Percentage Calculations

Percentage Formula:

$$\text{Percentage} = (\text{Count} \div \text{Total}) \times 100$$

Response	Count	Percentage (%)
Yes	110	55%
No	60	30%
Undecided	30	15%

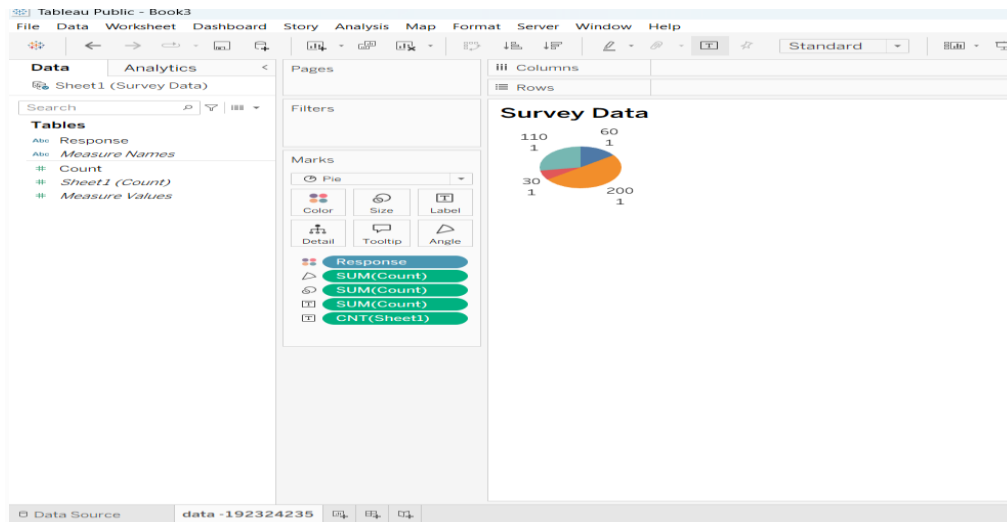
Angle Calculations

Angle Formula:

$$\text{Angle} = (\text{Percentage} \div 100) \times 360^\circ$$

Response	Percentage (%)	Calculation	Angle ($^\circ$)
Yes	55	$(55/100) \times 360$	198°
No	30	$(30/100) \times 360$	108°
Undecided	15	$(15/100) \times 360$	54°

$$\text{Total Angle} = 198^\circ + 108^\circ + 54^\circ = 360^\circ$$



Evidence

The pie chart indicates that Yes received the highest number of responses (110 participants, 55%, 198°). No accounts for 30% of the responses (108°), while Undecided represents the smallest group with 15% (54°).

Interpretation

The survey results show that more than half of the participants selected Yes, indicating strong overall agreement. The No responses form a significant minority, while relatively few participants remain undecided. The pie chart provides a quick visual comparison of the response distribution.

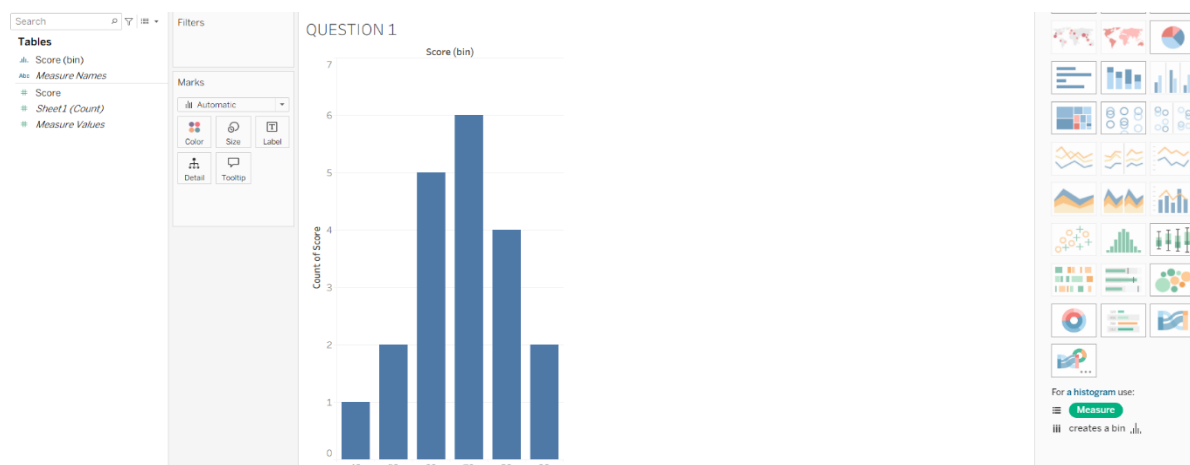
Conclusion

A pie chart is an effective method for displaying survey results because it converts raw counts into proportional slices of a circle. Converting frequencies into percentages and slice angles ensures an accurate representation of each response category and makes comparisons easy to understand.

Problem 9 – Histogram – Binning

Introduction

A histogram is a graphical representation of the frequency distribution of continuous numerical data. It groups data into equal-width intervals called **bins**, where the height of each bar represents the number of observations within that interval. Unlike a bar chart, the bars in a histogram touch each other because the data is continuous.



Data Range

- Lowest Score = **45**
- Highest Score = **95**
- Range = $95 - 45 = 50$
- Number of bins = **5**
- Bin Width = $50 \div 5 = 10$

Bin Boundaries and Frequencies

Bin	Scores Included	Frequency
45–54	45, 52	2
55–64	58, 61, 63	3
65–74	65, 67, 68, 70, 72, 73	6
75–84	75, 76, 78, 80, 82	5
85–95	85, 88, 90, 95	4

Evidence

The histogram shows that the highest concentration of students scored between **65 and 84 marks**. Very few students scored below 55, while four students scored above 85.

Interpretation

The score distribution indicates that most students achieved average to above-average marks. There are no major gaps between score ranges, suggesting a fairly balanced distribution without extreme clustering.

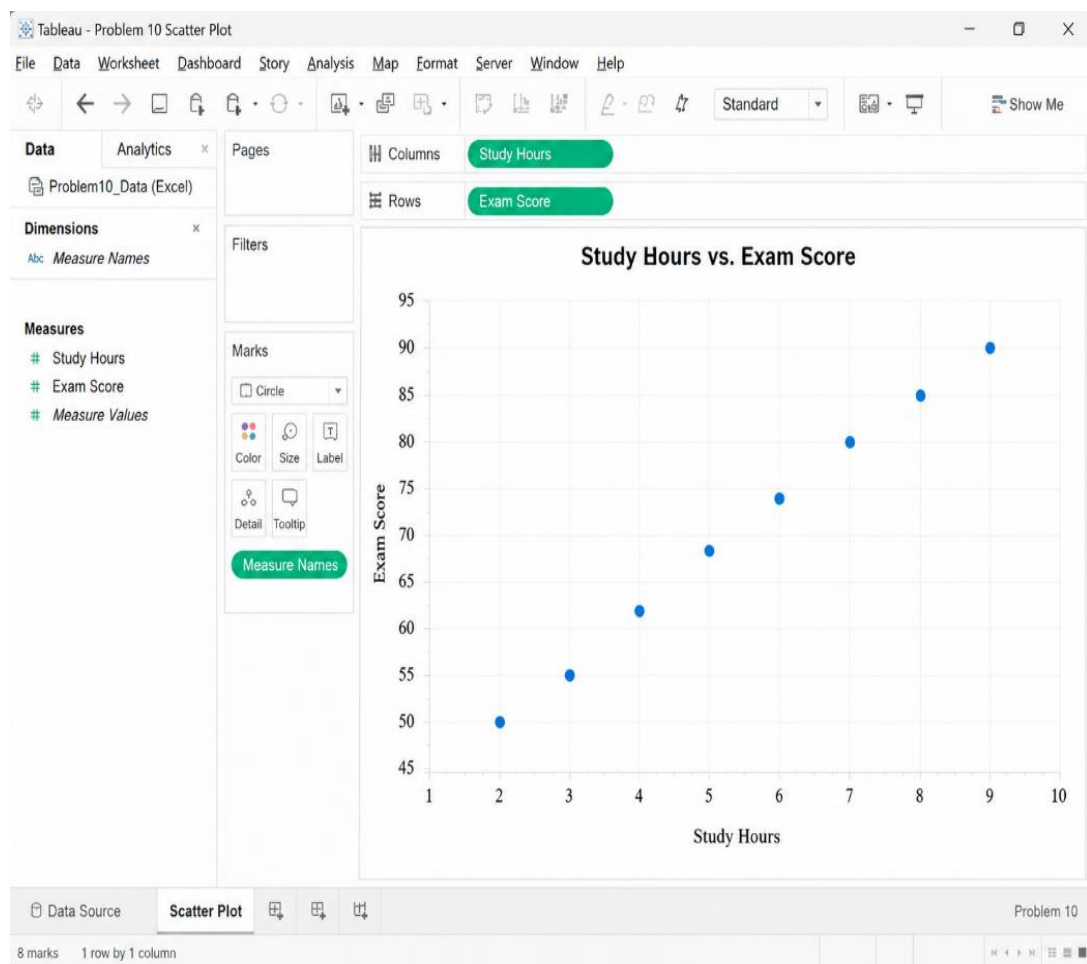
Conclusion

A histogram provides a clear visualization of how numerical data is distributed. It helps identify the most common score intervals, the spread of data, and overall performance patterns.

Problem 10 – Scatter Plot

Introduction

A scatter plot is used to study the relationship between two quantitative variables. Each observation is represented by a single point whose position is determined by its x-coordinate and y-coordinate.



- **X-axis:** Study Hours
- **Y-axis:** Exam Score

Each student is plotted individually.

Study Hours	Exam Score
2	50
3	55
4	62
5	68
6	74
7	80
8	85
9	90

Evidence

The plotted points gradually move upward from left to right without any major outliers. Students who studied more hours generally obtained higher scores.

Interpretation

The scatter plot indicates a **strong positive relationship** between study hours and exam scores. As study time increases, exam performance also increases. Although the scatter plot shows correlation, it does not prove that studying is the only factor affecting scores.

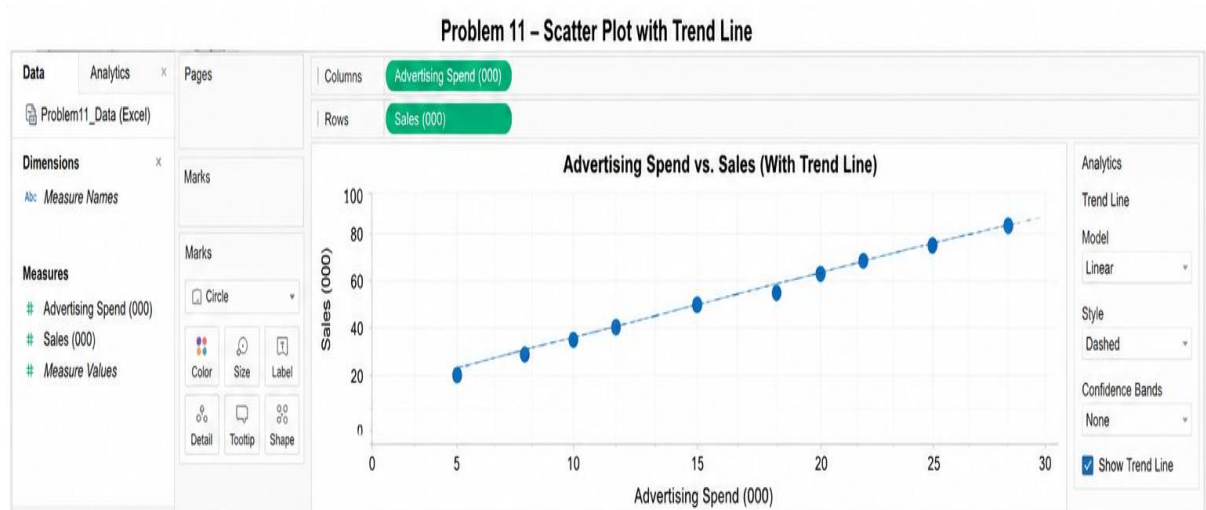
Conclusion

A scatter plot is an effective visualization for identifying relationships between two numerical variables. It helps detect trends, correlations, and unusual observations in the data.

Problem 11 – Scatter Plot with Trend Line

Introduction

A scatter plot with a trend line is used to visualize both the relationship between two variables and the overall direction of that relationship. The trend line summarizes the general pattern followed by the data points.



- **X-axis:** Advertising Spend (\$'000)
- **Y-axis:** Sales (\$'000)
- A linear trend line is drawn through the plotted points.

Evidence

Advertising Spend	Sales
5	20
8	28
10	35
12	40
15	50
18	55
20	62
22	68
25	75
28	82

The points lie close to an upward-sloping straight line.

Interpretation

The trend line indicates a **strong positive linear relationship**. Companies spending more on advertising generally achieve higher sales. Since the points lie close to the trend line, the relationship appears consistent with little variation.

Conclusion

A scatter plot with a trend line helps analyze the strength and direction of relationships between variables. It is widely used in business, economics, and marketing for prediction and decision-making.